15. d) +3

16. a) Both assertion and reason are true and reason is the correct explanation of assertion.

17. b) +3

18. a) Np, Pu ,Am

19. a) $La(OH)_2$ is less basic than $Lu(OH)_3$

UNIT-5

Choose the correct answer:

1. In the complex $\left[M(en)_2(Ox)\right]Cl$ For the central metal ion M^{3+}

   The primary valence is = +3

   The secondary valence = 6

   sum of primary valence and secondary valence = 3+6 = 9

   Answer : option (d)

2. The complex is $\left[M(H_2O)_5 Cl\right]Cl_2$

   1000 ml of 1M solution of the complex gives 2 moles of Cl^- ions

   1000 ml of 0.01M solution of the complex will give

   $$\frac{100 \text{ ml} \times 0.01\text{M} \times 2Cl^-}{1000 \text{ ml} \times 1\text{M}} = 0.002 \text{ moles of } Cl^- ions$$

   Answer : option (b)

3. Molecular formula: $MSO_4Cl.\ 6H_2O$.

   Formation of white precipitate with Barium chloride indicates that SO_4^{2-} ions are outside the coordination sphere, and no precipitate with $AgNO_3$ solution indicates that the Cl^- ions are inside the coordination sphere.Since the coordination number of M is 6, Cl^- and 5 H_2O are ligands, remaining 1 H_2O molecular and SO_4^{2-} are in the outer coordination sphere.

   Answer : option (c)

4. $\left[\overset{+}{Fe}(H_2O)_5\overset{+}{NO}\right]^{2+} SO_4^{2-}$

   +1 and +1 respectively

   Answer : option(d)

5. Answer : option(d)

6. Answer : option(d)

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